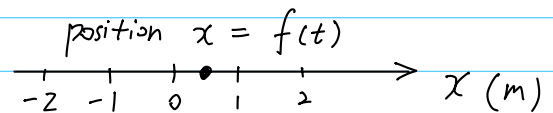


Review.

One-dimensional motion



- For a duration of time $\Delta t = t_2 - t_1$

displacement $\Delta x \equiv x_2 - x_1$ (unit: m) distance $d \equiv$ total length covered (m)

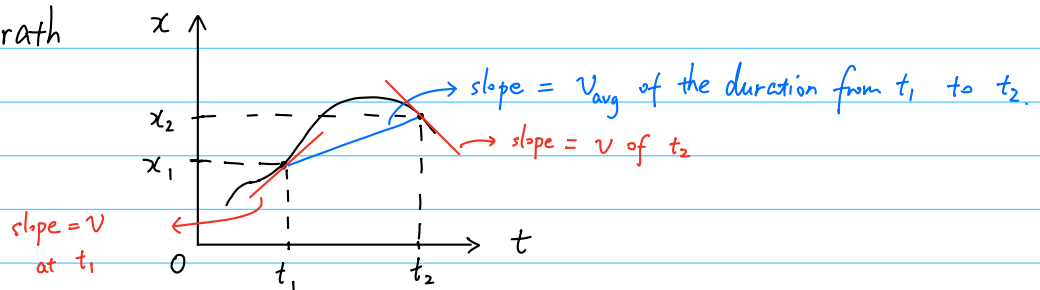
average velocity $v_{avg} \equiv \frac{\Delta x}{\Delta t}$ (m/s) average speed $\equiv \frac{d}{\Delta t}$ (m/s)

- At a specific moment

(instantaneous) velocity $v = \frac{dx}{dt}$ (m/s) instantaneous speed $= |v|$ (m/s)

"vector" or $x'(t)$ or $\dot{x}$ "scalar"

- $x-t$ graph



Q: At what moment(s) is v zero?

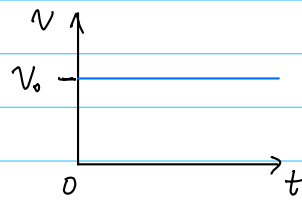
2.3 A special case: Particle under constant velocity

- both the magnitude & direction are constant (unchanged over time)

e.g. particle moves to the positive direction with constant speed v_0
Assume it passes through the origin at $t=0$.

Draw the velocity vs time, and position vs time diagrams.

$$v(t) = v_0$$



$$v = \frac{dx}{dt}$$

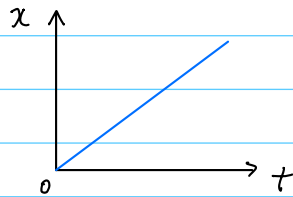
$$\Rightarrow x = \int v(t) dt$$

$$= \int v_0 dt$$

$$= v_0 t + \text{constant}$$

initial position

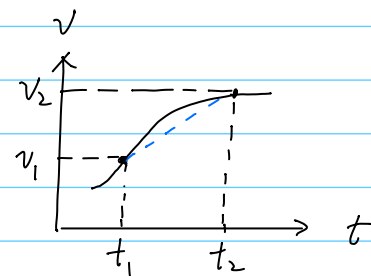
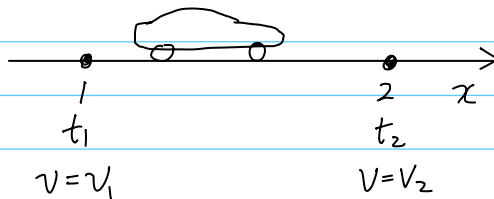
$$x = v_0 t + x_i$$



→ Q: what if the particle starts at $x = x_0$ instead of the origin?

* self-reading: 2.4 the analysis model approach to problem solving

2.5 If velocity changes with time? "object is accelerating"

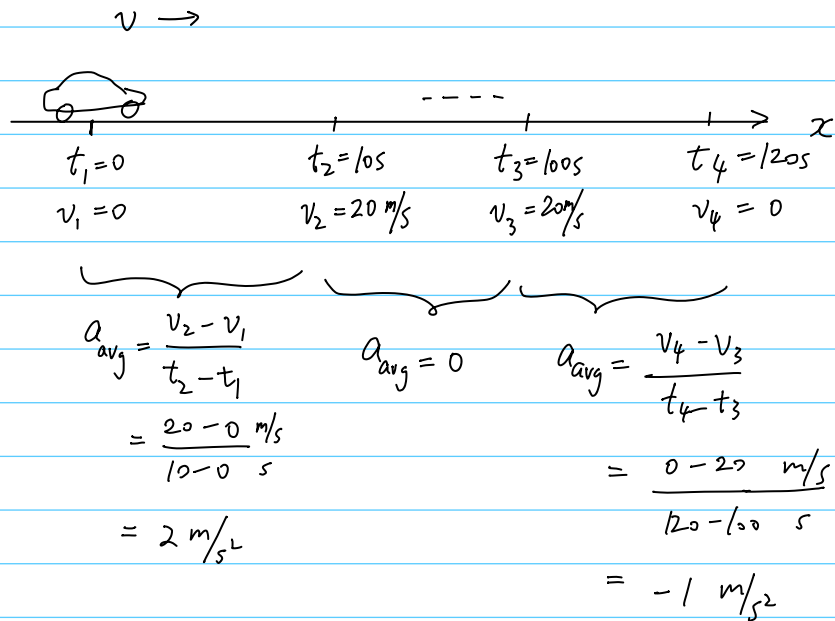


$$\text{average acceleration } a_{\text{avg}} \equiv \frac{\Delta v}{\Delta t} = \frac{v_2 - v_1}{t_2 - t_1}$$

= slope of dashed line

unit m/s^2

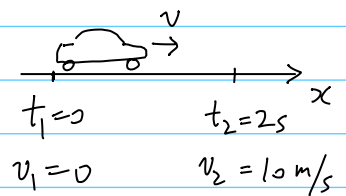
e.g.



acceleration is negative
 = decelerate in the positive direction
 = accelerate in the negative direction

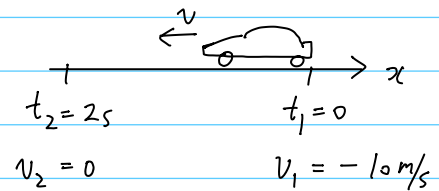
e.g.

accelerate toward $+x$



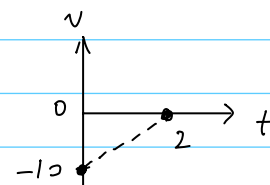
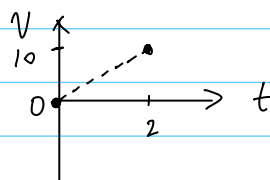
$$a_{avg} = \frac{10 - 0}{2 - 0} = 5 \text{ m/s}^2$$

decelerate toward $-x$



$$a_{avg} = \frac{0 - (-10)}{2 - 0} = 5 \text{ m/s}^2$$

↓ same acceleration



What if $\Delta t \rightarrow 0$?

instantaneous acceleration

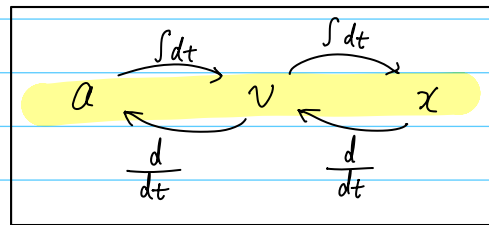
$$a(t) \equiv \lim_{\Delta t \rightarrow 0} \frac{\Delta v}{\Delta t} = \frac{dv}{dt} = \frac{d}{dt} \left(\frac{dx}{dt} \right) = \frac{d^2 x}{dt^2}$$

or $x''(t)$ or $\ddot{x}$

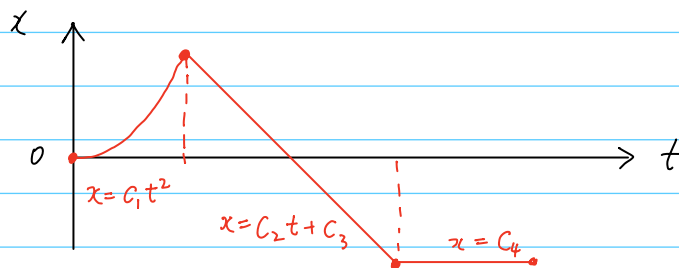
For a period of time from t_1 to t_2 :

$$\Delta v = \int_{t_1}^{t_2} a(t) dt$$

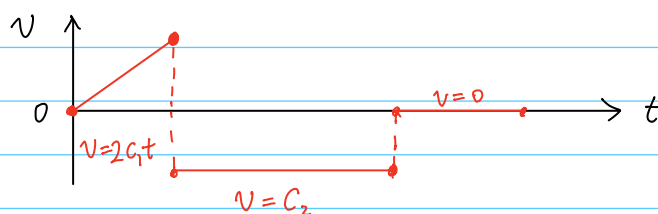
$$\Delta x = \int_{t_1}^{t_2} v(t) dt$$



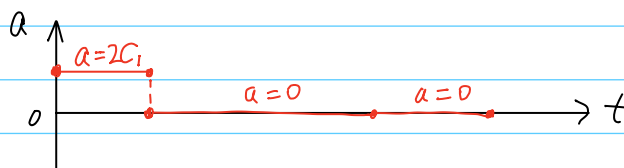
e.g. Given $x-t$ relations, find $v-t$ and $a-t$ relations.



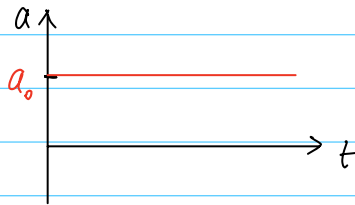
(c_1 to c_4 are constants)



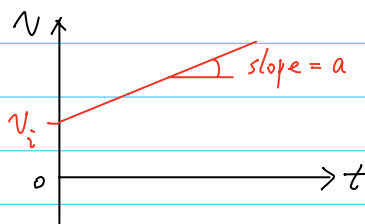
math ref: $\frac{d t^n}{dt} = n t^{n-1}$



2.7

model: particle under constant acceleration

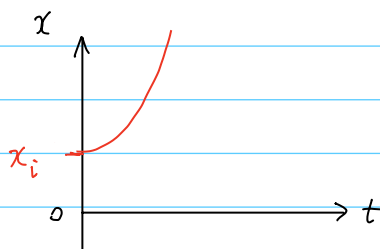
$$a(t) = a_0 = \text{constant}$$

1. From a to v :

$$v(t) = \int a_0 dt = a_0 t + v_i \quad (1)$$

math. ref:

$$\text{indefinite integral } \int t^n dt = \frac{1}{n+1} t^{n+1} + \text{constant}$$

when $t=0$, $v(t) = v_i$ (initial velocity)2. From v to x :

$$x(t) = \int v(t) dt$$

$$= \int (a_0 t + v_i) dt$$

$$= \frac{a_0 t^2}{2} + v_i t + x_i \quad (2)$$

when $t=0$, $x(t) = x_i$ (initial position)

Three other useful equations derived from (1) & (2):

Express x_f with v_f and v_i without a :

$$\begin{aligned}\text{From (1): } a_0 &= \frac{v_f - v_i}{t}, \text{ put in (2): } x_f = \frac{1}{2} \cdot \frac{v_f - v_i}{t} \cdot t^2 + v_i t + x_i \\ &= \left(\frac{1}{2} v_f - \frac{1}{2} v_i + v_i \right) t + x_i\end{aligned}$$

$$\boxed{x_f = \frac{1}{2} (v_f + v_i) t + x_i} \quad (3)$$

average velocity (from $t_i=0$ to t_f):

$$v_{\text{avg}} = \frac{\Delta x}{\Delta t} = \frac{x_f - x_i}{t_f - t_i}$$

$$\text{From (2), } x(t=0) = x_i$$

$$x(t=t_f) = \frac{1}{2} a_0 t_f^2 + v_i t_f + x_i$$

$$\therefore v_{\text{avg}} = \frac{\frac{1}{2} a_0 t_f^2 + v_i t_f + \cancel{x_i} - \cancel{x_i}}{t_f}$$

$$= \frac{1}{2} a_0 t_f + v_i$$

$$\text{From (1), } v_f = a_0 t_f + v_i \Rightarrow a_0 t_f = v_f - v_i$$

$$\therefore v_{\text{avg}} = \frac{1}{2} (v_f - v_i) + v_i = \boxed{\frac{1}{2} (v_f + v_i)} \quad (4)$$

For constant acceleration motion, average velocity of a trip is the numerical average of the initial and final velocities.

Express v_f with x and a without t :

$$\text{From (1)} \quad v_f = a_0 t + v_i$$

$$\Rightarrow t = \frac{v_f - v_i}{a_0}$$

Plug into (2):

$$x_f = \frac{1}{2} a_0 t^2 + v_i t + x_i$$

$$= \frac{1}{2} a_0 \cdot \frac{(v_f - v_i)^2}{a_0^2} + v_i \cdot \frac{v_f - v_i}{a_0} + x_i$$

$$= \frac{1}{2a_0} (v_f^2 - 2v_f v_i + v_i^2) + \frac{1}{a_0} (v_f v_i - v_i^2) + x_i$$

$$2a_0 x_f = v_f^2 - \cancel{2v_f v_i} + v_i^2 + \cancel{2v_f v_i} - 2v_i^2 + 2a_0 x_i$$

$$\boxed{2a_0(x_f - x_i) = v_f^2 - v_i^2}$$

(5)

$$\text{or } \boxed{v_f^2 = v_i^2 + 2a_0(x_f - x_i)}$$

This is convenient when problem does not involve t .

★ Eq. (1) to (5) are only for constant acceleration motion.

Note: constant velocity motion is a special case of constant acceleration motion where $a_0 = 0$

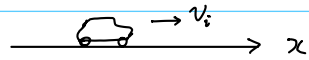
e.g. put $a_0 = 0$ into (1) & (2) we get

$$v(t) = v_i$$

$$x(t) = v_i t + x_i$$

This goes back to the model described in session 2.3.

e.g. At the initial moment $t=0$, v is v_i along $+x$ direction.

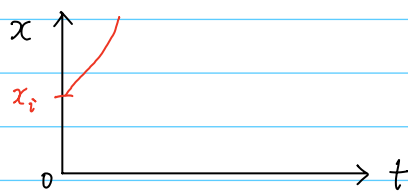
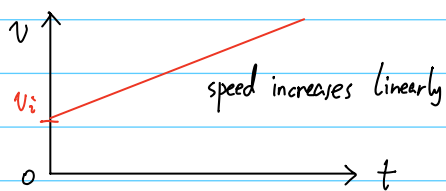
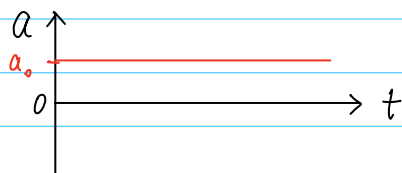


if: a) $a = a_0 > 0$

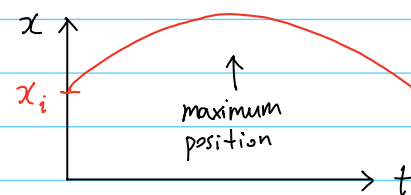
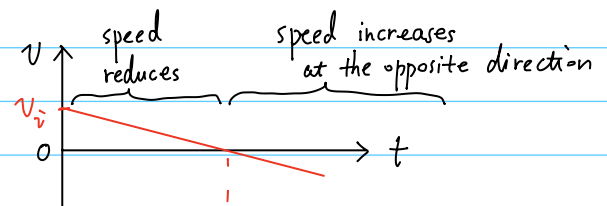
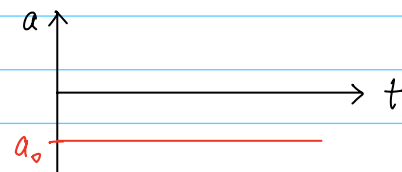
b) $a = a_0 < 0$,

How do $v-t$ and $x-t$ diagrams look like?

a)



b)



2.8. Free falling objects

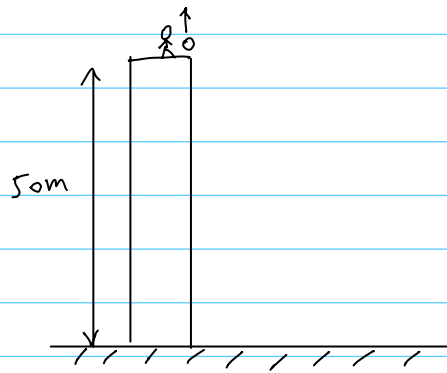
Definition: when an object near Earth's surface only experiences gravity force.
(when air friction is negligible.)

* The total force exerted on an object generates an acceleration.

Earth's gravitational acceleration is constant (almost).

$g \approx 9.8 \text{ m/s}^2$,
pointing towards ground.

e.g. Person throws ball upwards from the top of a 50 m tall building,
ball initial speed 20 m/s. Use $g = 9.8 \text{ m/s}^2$.



(a) time to reach maximum height?

(b) maximum height (from ground)?

(c) final velocity when hits the ground?

(d) time when hits the ground?

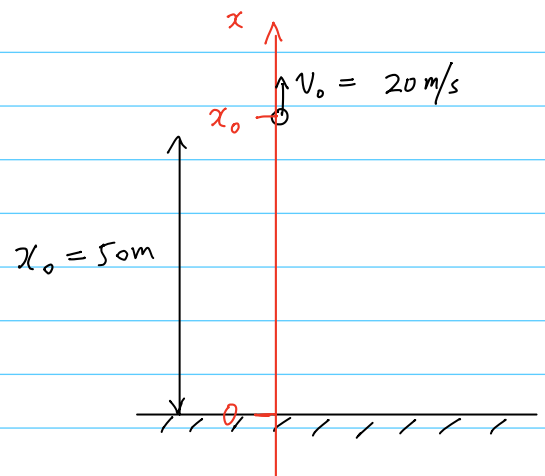
1. Set up coordinates.

Mark specific time points:

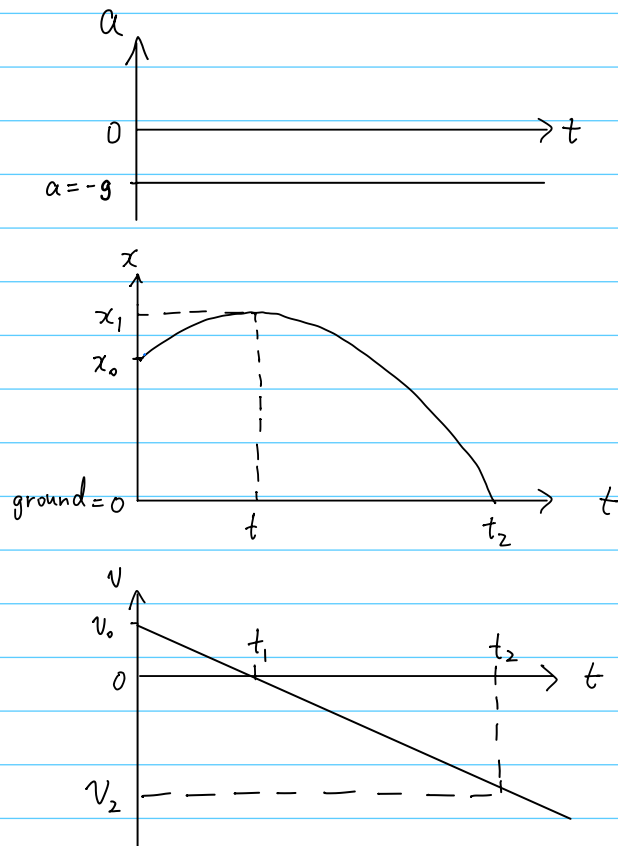
t_0 — starts from top of building. Let be zero.

t_1 — reaches max height

t_2 — hits ground



2. Analyze with diagrams.



Analyze what are the given quantities and what we need to find out.

x_0 , v_0 , a (which is $-g$),

t_1 , x_1 , v_2 , t_2

3. Decide which time duration you will analyze and write down equations for that duration.

4. Solve equations. (use known quantities to express unknown.)

(a) time to reach maximum height?

Consider from t_0 to t_1

	initial	final	
t	$t_0 = 0$	t_1	
x	$x_0 = 50 \text{ m/s}$	x_1	
v	$v_0 = 20 \text{ m/s}$	$v_1 = 0$	★ Hidden condition: v at maximum height is 0.
a	$a = -g = -9.8 \text{ m/s}^2$		

Use eq.(1) $\underline{v(t)} = \underline{a}t + \underline{v_i}$ use symbols

$$v_1 = -g t_1 + v_0$$

$$t_1 = \frac{v_1 - v_0}{-g} = \frac{0 - 20}{-9.8} \approx 2.04 \text{ s}$$

replace with numerics.

(b) maximum height (from ground)?

Use eq.(2) $\underline{x(t)} = \frac{1}{2} \underline{a}t^2 + \underline{v_i}t + \underline{x_i}$

$$x_1 = -\frac{1}{2} g t_1^2 + v_0 t_1 + x_0$$

$$= \frac{1}{2} \times (-9.8) \times 2.04^2 + 20 \times 2.04 + 50$$

$$\approx 70.4 \text{ m}$$

(c) final velocity when hits the ground?

Consider from t_0 to t_2

	initial	final
t	$t_0 = 0$	t_2
x	$x_0 = 50 \text{ m}$	$x_2 = 0$
v	$v_0 = 20 \text{ m/s}$	v_2
a	$a = -g = -9.8 \text{ m/s}^2$	

From eq. (5) $\underline{v_f^2} = \underline{v_i^2} + \underline{2a}(\underline{x_f} - \underline{x_i})$

$$v_2^2 = v_0^2 - 2g(x_2 - x_0)$$

$$v_2 = \sqrt{v_0^2 - 2g(x_2 - x_0)}$$

$$= \sqrt{20^2 - 2 \times 9.8 \times (0 - 50)}$$

$$\approx \pm 37.1 \text{ m/s}$$

★ After square root there can be two possibilities + or -

here v is pointing downwards, so $v_2 = -37.1 \text{ m/s}$

* check dimensions of the final answer $\sqrt{v_0^2 + 2gx_0}$

$$\sqrt{\left(\frac{\text{m}}{\text{s}}\right)^2 - \frac{\text{m}}{\text{s}^2} \cdot \text{m}}$$

$$= \sqrt{\frac{\text{m}^2}{\text{s}^2} - \frac{\text{m}^2}{\text{s}^2}}$$

$$= \text{m/s} \Rightarrow \text{unit of velocity } \checkmark$$

(d) time when hits the ground?

Method 1: Consider from t_0 to t_2

	initial	final
t	$t_0 = 0$	t_2
x	$x_0 = 50 \text{ m/s}$	$x_2 = 0$
v	$v_0 = 20 \text{ m/s}$	$v_2 \leftarrow v_2 = -37.1 \text{ m/s}$
a	$a = -g = -9.8 \text{ m/s}^2$	

From eq. (1) $\underline{v_f} = \underline{a}t + \underline{v_i}$

$$v_2 = -g t_2 + v_0$$

$$\begin{aligned} t_2 &= \frac{v_2 - v_0}{-g} \\ &= \frac{-37.1 - 20}{-9.8} \\ &\approx 5.8 \text{ s} \end{aligned}$$

Method 2 Consider from t_0 to t_2

From eq. (2) $\underline{x_f} = \frac{1}{2} \underline{a} t^2 + \underline{v_i} t + \underline{x_i}$

$$x_2 = -\frac{1}{2} g t_2^2 + v_0 t_2 + x_0$$

math ref:

$$a x^2 + b x + c = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

We have $\frac{1}{2} \times (-9.8) \times t^2 + 20t + 50 = 0$

Solve this quadratic equation

$$x \approx -1.75 \text{ or } \underline{5.83 \text{ s}}$$

other methods: You can also use the duration from t_1 to t_2 .

Be aware $(t_2 - t_1)$ should be used for t .